

### **3. Log Analysis using Splunk**

#### **→ STEP 1 — INSTALL SPLUNK FREE**

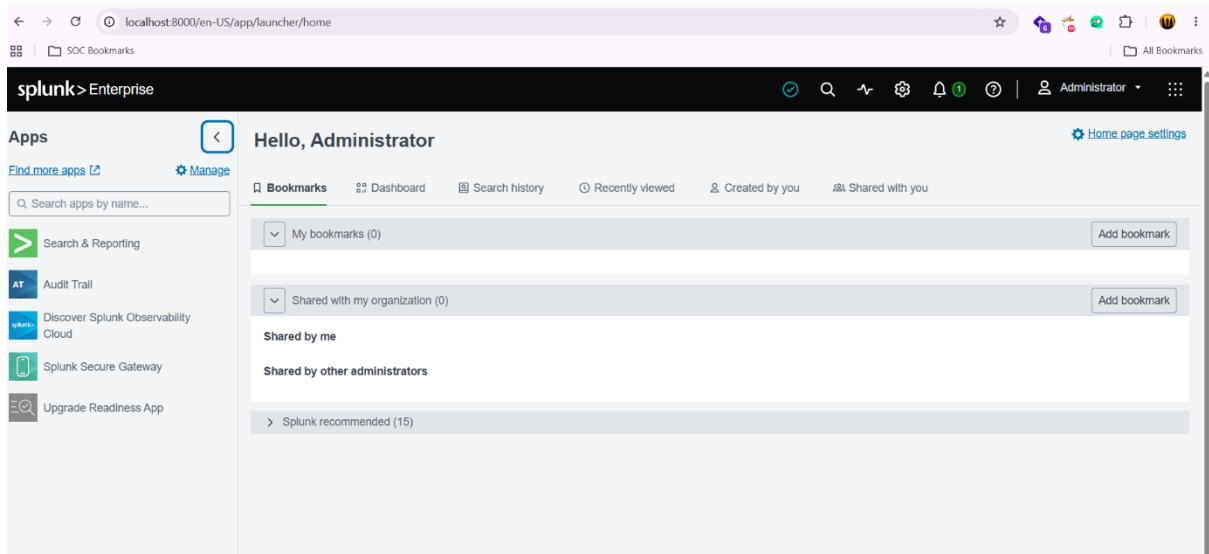
##### **1.1 — Download Splunk Free**

###### **Exact Action:**

1. Open your web browser (Chrome/Edge/Firefox)
2. Go  
to: [https://www.splunk.com/en\\_us/download/splunk-enterprise.html](https://www.splunk.com/en_us/download/splunk-enterprise.html)
3. Click the "**Free Splunk**" or "**Get My Free Splunk**" button (you'll need to create a free account)
4. On the download page, choose **Splunk Enterprise** (the free license gives you 500MB/day — plenty for this lab)
5. **Pick the right version:**
  - If you're on **Windows 10/11**: Choose the **64-bit .msi installer** (e.g., splunk-9.x.x-xxxxx-x64-release.msi)

## → First Login — Splunk Home Dashboard

1. Open your browser
2. Navigate to: <http://localhost:8000>
3. Login with:
  - Username: **admin**
  - Password: **Changeme1!**



You should see the **Splunk Home Dashboard** with :

- A search bar at the top (white bar with "Search here..." placeholder)
- Left Side Bar with Options : Apps, Search Reporting, Data Summary
- Main area showing:
  - Search & Reporting
  - Add Data
  - Dashboards
  - Settings

**Common tasks** Hide for users

|                                                                                                                                                                                  |                                                                                                                                                        |                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Add data</b><br>Add data from a variety of common sources.                                  |  <b>Search your data</b><br>Turn data into doing with Splunk search.  |  <b>Visualize your data</b><br>Create dashboards that work for your data. |
|  <b>Manage alerts</b><br>Manage the alerts that monitor your data.                              |  <b>Add team members</b><br>Add your team members to Splunk platform. |  <b>Manage permissions</b><br>Control who has access with roles.          |
|  <b>Configure mobile devices</b><br>Login or manage mobile devices using Splunk Secure Gateway. |                                                                                                                                                        |                                                                                                                                                              |

**Learning & resources** Hide for users

|                                                                                                                                                                                |                                                                                                                                                                                                                |                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Product tours</b><br>New to Splunk? Take a tour to help you on your way.                  |  <b>Learn more with Splunk Docs</b> <a href="#">↗</a><br>Deploy, manage, and use Splunk software with comprehensive guidance. |  <b>Get help from Splunk experts</b> <a href="#">↗</a><br>Actionable guidance on the Splunk Lantern Customer Success Center. |
|  <b>Extend your capabilities</b> <a href="#">↗</a><br>Browse thousands of apps on Splunkbase. |  <b>Join the Splunk Community</b> <a href="#">↗</a><br>Learn, get inspired, and share knowledge.                              |  <b>See how others use Splunk</b> <a href="#">↗</a><br>Browse real customer stories.                                         |

## → GET LOGS

### Get Logs From NirSoft Tools

1. Open your browser and go to: [https://www.nirsoft.net/utls/computer\\_activity\\_view.html](https://www.nirsoft.net/utls/computer_activity_view.html)
2. Scroll down and click "Download LastActivityView" (or "Download LastActivityView (64-bit)")
3. Extract the ZIP file to a folder (e.g., C:\nirsoft\)
4. Double-click LastActivityView.exe

### Disclaimer

The software is provided "AS IS" without any warranty, either expressed or implied, including, but not limited to, the implied warranties of merchantability and fitness for a particular purpose. The author will not be liable for any special, incidental, consequential or indirect damages due to loss of data or any other reason.

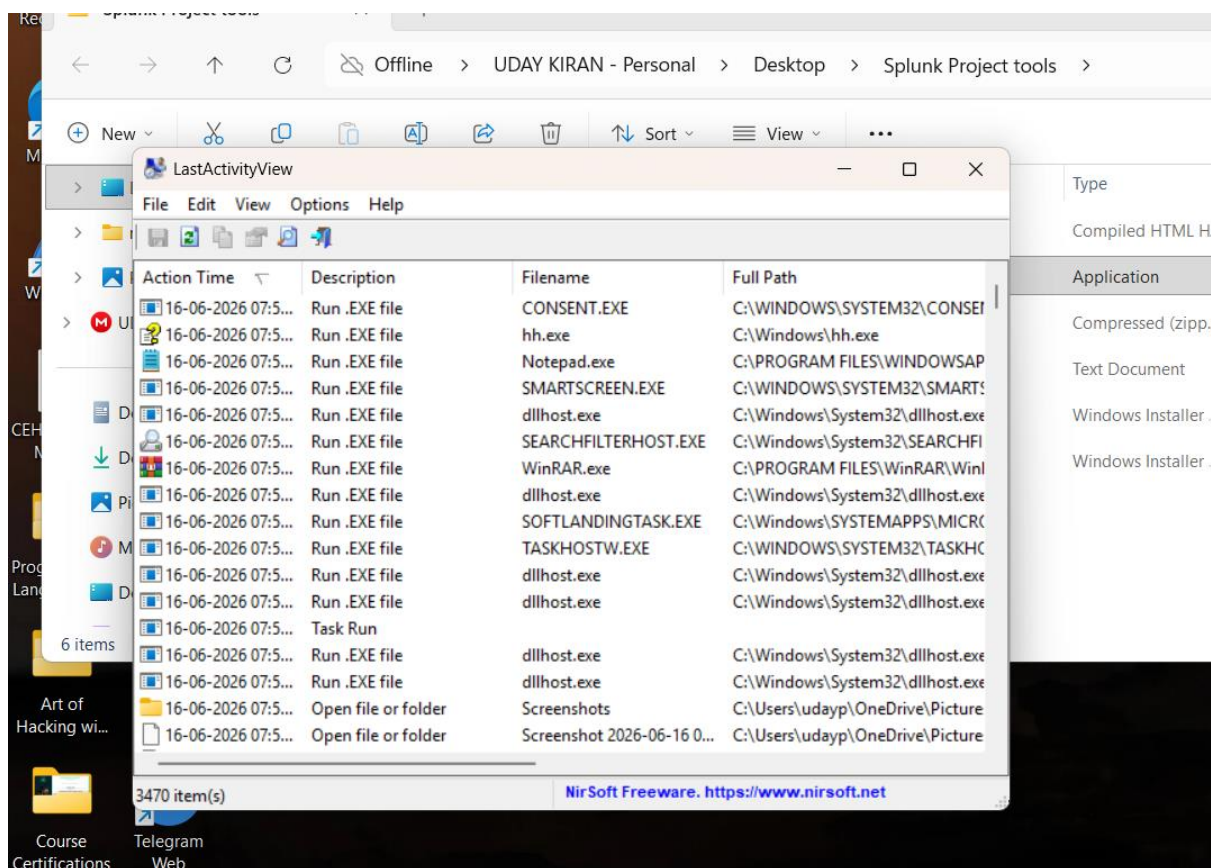
### Feedback

If you have any problem, suggestion, comment, or you found a bug in my utility, you can send a message to [nirsofer@yahoo.com](mailto:nirsofer@yahoo.com)

[Download LastActivityView](#)

LastActivityView is also available in other languages. In order to change the language of LastActivityView, download the appropriate language zip file, extract the 'lastactivityview\_lng.ini', and put it in the same folder that you installed LastActivityView utility.

## Click On Download “LastActivityView”

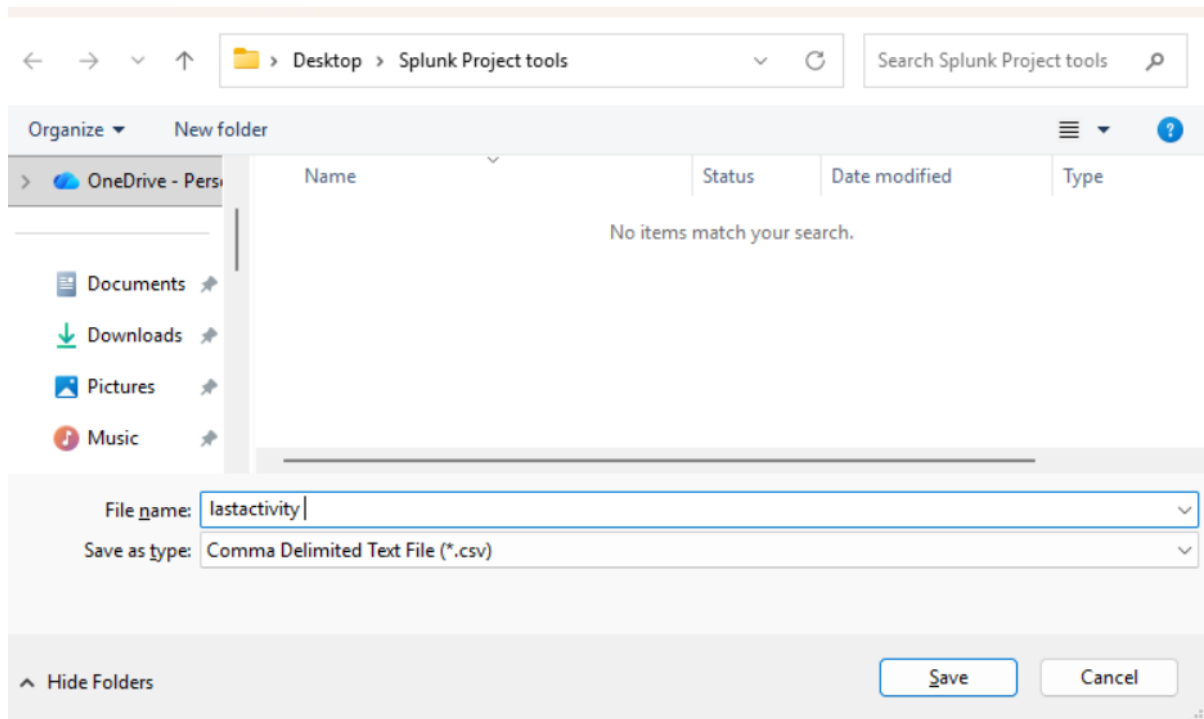


You'll see a window with columns like:

- **Date/Time**
- **Activity Type** (Logon, Logoff, ScreenSaver, etc.)
- **Description**
- **Filename/Process**

## Export to CSV:

1. Click **File** → **Save All Items** (Ctrl+A, Ctrl+S)
2. Choose **"Tab separated file / CSV"** from the dropdown
3. Save as: **lastactivity.csv**



If you open This Saved File You see all Last Activities.

|                                                                                                                                                                                                           |             |                                                              |                |                 |                                                                                            |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------|----------------|-----------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|------------------------------------------------------------------------------------------------------------------------|---|---|--|--|
| 1 POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As... |             |                                                              |                |                 |                                                                                            |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| A1                                                                                                                                                                                                        |             |                                                              |                |                 |                                                                                            |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
|                                                                                                                                                                                                           | A           | B                                                            | C              | D               | E                                                                                          | F                                                     | G                   | H | I | J | K | L | M | N | O | P | Q | R | S | T | U                                                                                                                      | V | W |  |  |
| 1                                                                                                                                                                                                         | Action      | Description                                                  | Filename       | Full Path       | More Info                                                                                  | File                                                  | Externs Data Source |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 2                                                                                                                                                                                                         | Run .EXE    | f CONSENT                                                    | C:\WINDOWS     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\CONSENT.EXE-40419367.pf           |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 3                                                                                                                                                                                                         | Run .EXE    | f hh.exe                                                     | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\HH.EXE-353EF707.pf                |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 4                                                                                                                                                                                                         | Run .EXE    | f Notepad.e                                                  | C:\PROGR       | ,, ,            | exe                                                                                        | C:\WINDOWS\Prefetch\NOTEPAD.EXE-22B0553.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 5                                                                                                                                                                                                         | Run .EXE    | f SMARTSCF                                                   | C:\WINDOWS     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\SMARTSCREEN.EXE-EACC1250.pf       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 6                                                                                                                                                                                                         | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-705CE0CA.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 7                                                                                                                                                                                                         | Run .EXE    | f SEARCHFIL                                                  | C:\Windows     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\SEARCHFILTERHOST.EXE-44162447.pf  |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 8                                                                                                                                                                                                         | Run .EXE    | f WinRAR.e                                                   | C:\PROGR       | Alexander       | exe                                                                                        | C:\WINDOWS\Prefetch\WINRAR.EXE-AEA955E.pf             |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 9                                                                                                                                                                                                         | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-92A71551.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 10                                                                                                                                                                                                        | Run .EXE    | f SOFTLAND                                                   | C:\Windows     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\SOFTLANDINGTASK.EXE-A93C9879.pf   |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 11                                                                                                                                                                                                        | Run .EXE    | f TASKHOST                                                   | C:\WINDOWS     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\TASKHOSTW.EXE-2E5D4B75.pf         |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 12                                                                                                                                                                                                        | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-895818BD.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 13                                                                                                                                                                                                        | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-841980A1.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 14                                                                                                                                                                                                        | Task Run    | SoftLandingDeferralTask-{cbe79e42-b6ef-4367-89ed-49d545c339} |                |                 |                                                                                            |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   | SoftLanding\S-1-5-21-3883017404-3009620963-277394729-1001\SoftLandingDeferralTask-{cbe79e42-b6ef-4367-89ed-49d545c339} |   |   |  |  |
| 15                                                                                                                                                                                                        | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-691D258.pf             |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 16                                                                                                                                                                                                        | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-705CE0CA.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 17                                                                                                                                                                                                        | Open file   | c:\Screensho                                                 | C:\Users\udayp | OneDrive\Pictur | C:\Users\udayp\AppData\Roaming\Microsoft\Windows\Recent\Screenshots (7).lnk                |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 18                                                                                                                                                                                                        | Open file   | c:\Screensho                                                 | C:\Users\udayp | One\ng          | C:\Users\udayp\AppData\Roaming\Microsoft\Windows\Recent\Screenshots 2026-06-16 0705305.lnk |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 19                                                                                                                                                                                                        | Select file | c:\Screensho                                                 | C:\Users\udayp | One\ng          | C:\Users\udayp\AppData\Roaming\Microsoft\Windows\Recent\Screenshots 2026-06-16 0705305.lnk |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 20                                                                                                                                                                                                        | Select file | c:\Screensho                                                 | C:\Users\udayp | One\ng          | C:\Users\udayp\AppData\Roaming\Microsoft\Windows\Recent\Screenshots 2026-06-16 0705305.lnk |                                                       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 21                                                                                                                                                                                                        | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-705CE0CA.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 22                                                                                                                                                                                                        | Run .EXE    | f dlhost.exe                                                 | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\DLHOST.EXE-705CE0CA.pf            |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 23                                                                                                                                                                                                        | Run .EXE    | f SEARCHFIL                                                  | C:\Windows     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\SEARCHFILTERHOST.EXE-44162447.pf  |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 24                                                                                                                                                                                                        | Run .EXE    | f SEARCHFIL                                                  | C:\Windows     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\SEARCHFILTERHOST.EXE-44162447.pf  |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 25                                                                                                                                                                                                        | Run .EXE    | f SEARCHPR                                                   | C:\Windows     | Microsoft       | EXE                                                                                        | C:\WINDOWS\Prefetch\SEARCHPROTOCOLHOST.EXE-69AC563.pf |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |
| 26                                                                                                                                                                                                        | Run .EXE    | f svchost.e                                                  | C:\Windows     | Microsoft       | exe                                                                                        | C:\WINDOWS\Prefetch\SVCHOST.EXE-18C5C66A.pf           |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                        |   |   |  |  |

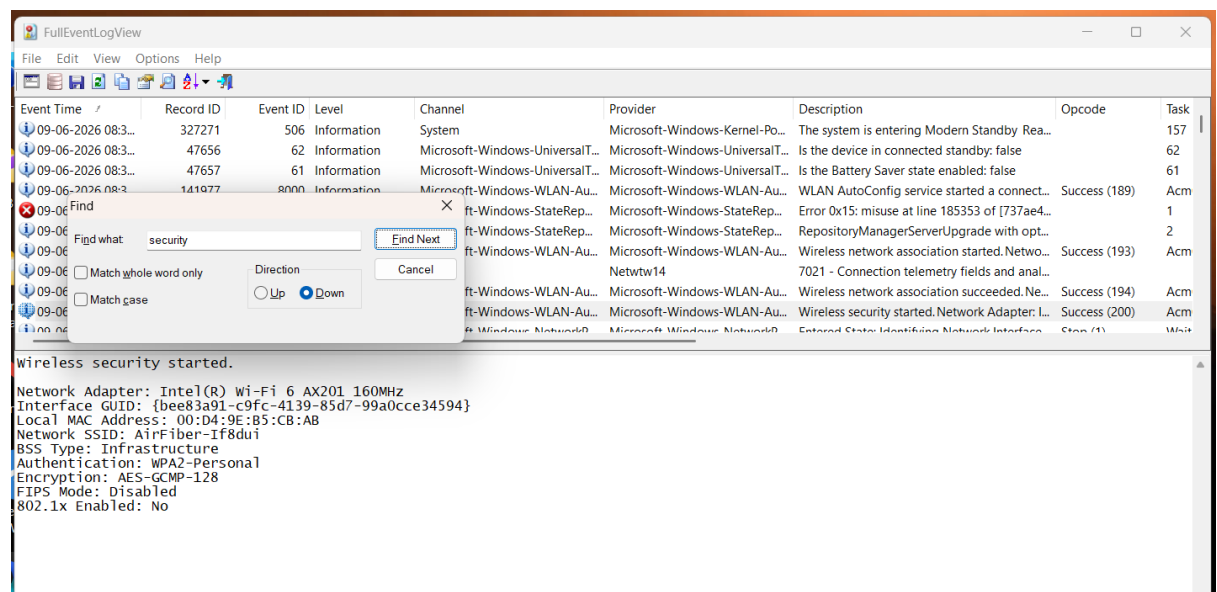
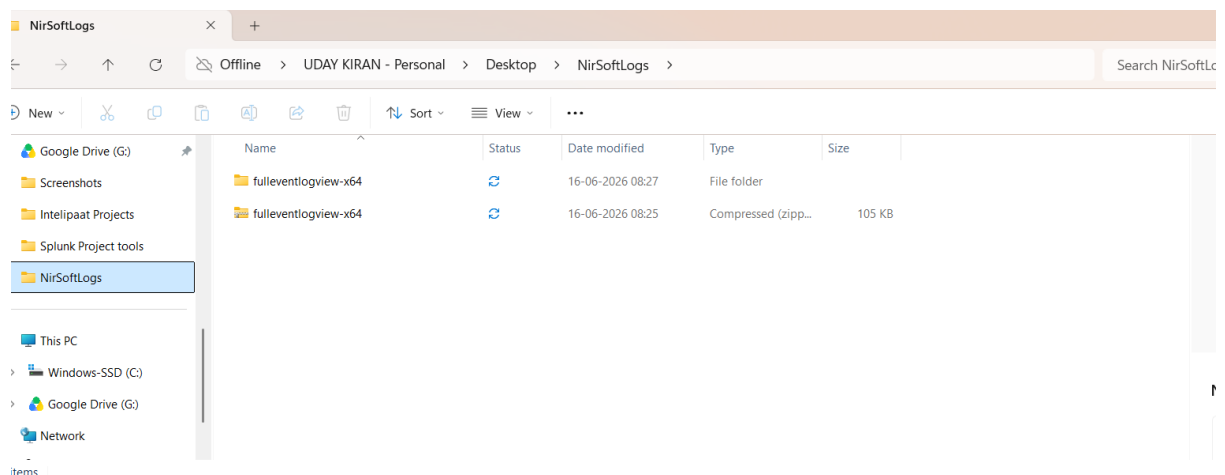
# Download FullEventLogView (Better for Security Logs)

1. Goto

[https://www.nirsoft.net/utils/full\\_event\\_log\\_view.html](https://www.nirsoft.net/utils/full_event_log_view.html)

2. Download full **Eventlog View** (64-bit recommended)

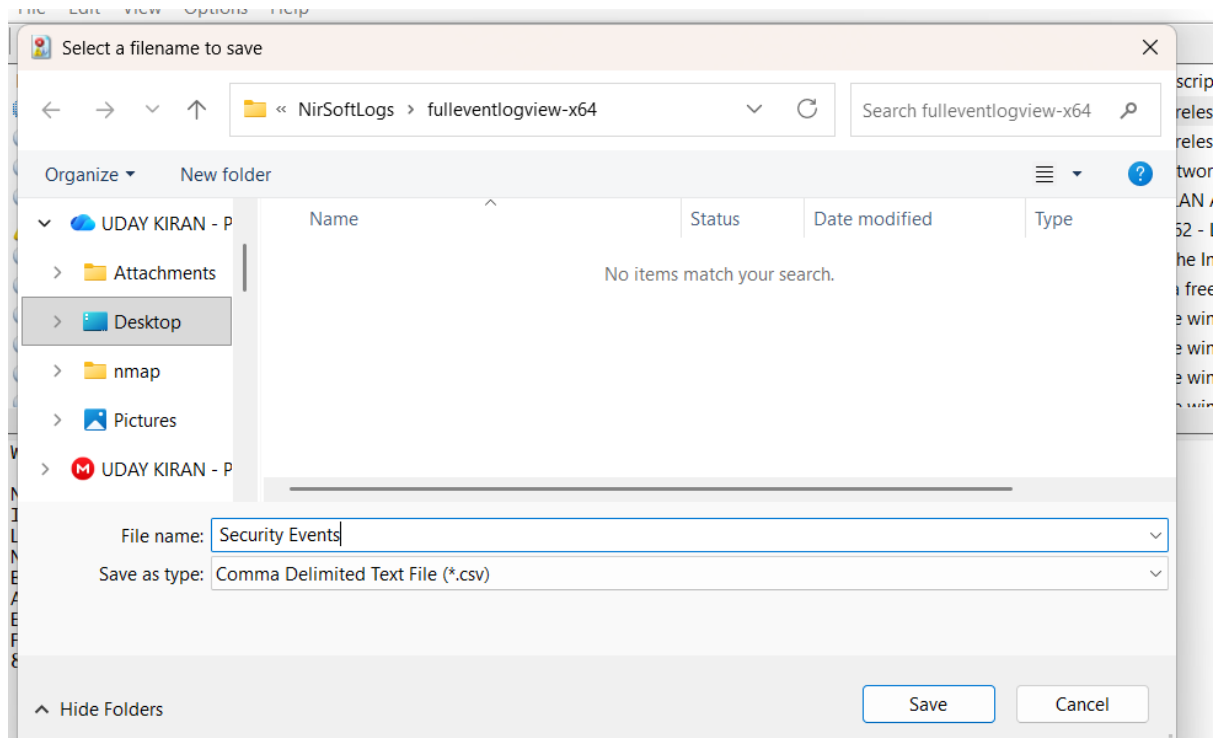
3. Extract and run the .exe



1. In FullEventView, find "**Security**" in the list

2. Click it → Click "**View Log Data**" at the top

3. Or: Right-click **Security** → "**Save Selected Items**"



Click On Save

↑ ↻ Offline > UDAY KIRAN - Personal > Desktop > NirSoftLogs > fulleventlogview-x64

| Name             | Status | Date modified    | Type                  | Size   |
|------------------|--------|------------------|-----------------------|--------|
| FullEventLogView |        | 16-06-2026 08:43 | CFG File              | 2 KB   |
| FullEventLogView |        | 16-06-2026 08:27 | Compiled HTML H...    | 21 KB  |
| FullEventLogView |        | 16-06-2026 08:27 | Application           | 179 KB |
| readme           |        | 16-06-2026 08:27 | Text Document         | 20 KB  |
| Security Events  |        | 16-06-2026 08:48 | Microsoft Excel Co... | 1 KB   |

A table showing Windows Event Log entries with columns like

- **Event ID** (4624, 4625, 4648, etc.)
- **Date/Time**
- **Source**
- **Message** (contains usernames, IPs, etc.)

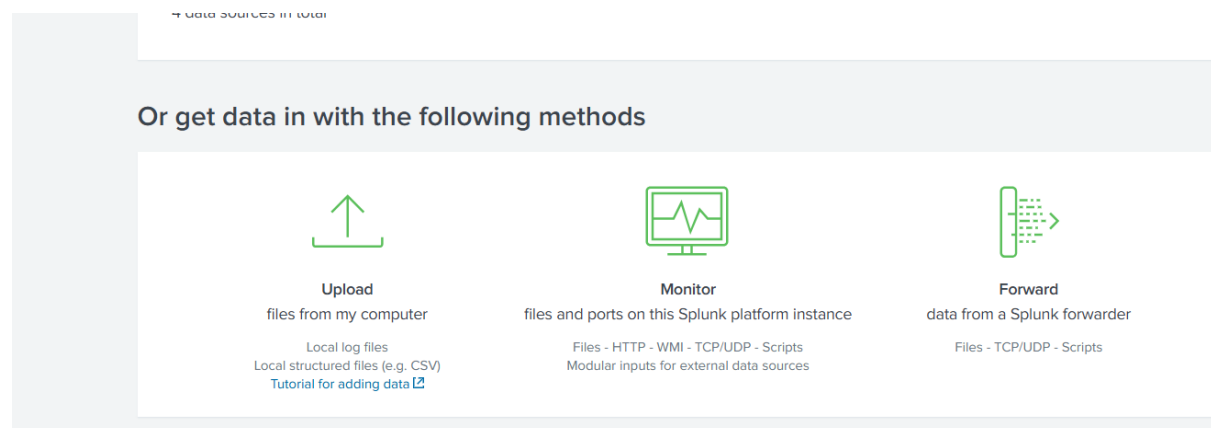
## Key fields in the message text:

- EventID 4625 → Failed login (this is our target!)
- EventID 4624 → Successful login
- EventID 4648 → Explicit credential usage
- Look for: Account Name, Source Network Address, Logon Type

**Why this step matters:** Real Windows Security logs contain actual failed login attempts from your system, making the analysis authentic and immediately relevant to your environment.

## INGEST LOGS INTO SPLUNK

Open Splunk and Click on Add data



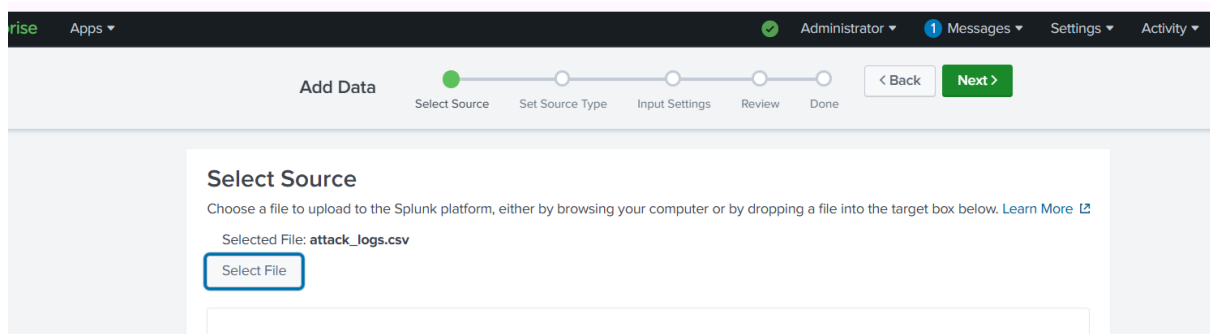
You'll see the **Add Data** page with 3 steps at the top in a horizontal wizard bar:

Step 1: Select Source → Step 2: Set Source Type → Step 3: Input Settings → Step 4: Review

And three large icons below:

- **Upload** (cloud with up-arrow) ← Click this
- **Monitor** (folder with eye)
- **Forward** (network arrows)

Click On upload → The Add Data wizard is how you get external log files into Splunk's indexing engine so they become searchable.



You'll see a **File Preview Table** showing:

- A sample of your CSV data rendered as a table

## SET Source type

1. Click "Next" or go to Step 2: Set Source Type
2. In the "Source type" dropdown:
  - Start typing "csv" → Select csv from the list

## Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **attack\_logs.csv**

 Job terminated unexpectedly

Source type: csv ▾

Save As

> Timestamp

> Delimited settings

> Advanced

The source type dropdown with "csv" highlighted, and below it the file preview showing the correctly parsed columns.

→ Setting the correct source type tells Splunk how to parse the data. CSV tells Splunk to treat commas as field separators, making each column a searchable field.

## Input Settings → Create New Index

|                 |                           |
|-----------------|---------------------------|
| Index Name      | security_lab              |
| Index Data Type | Events (default)          |
| Max Size        | 100 MB (or leave default) |

Click On Save

**New Index**

**General Settings**

Index Name   
Set index name (e.g., INDEX\_NAME). Search using index=INDEX\_NAME.

Index Data Type ☒ Events ☐ Metrics  
The type of data to store (event-based or metrics).

Home Path optional  
Hot/warm db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/db).

Cold Path optional  
Cold db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/colddb).

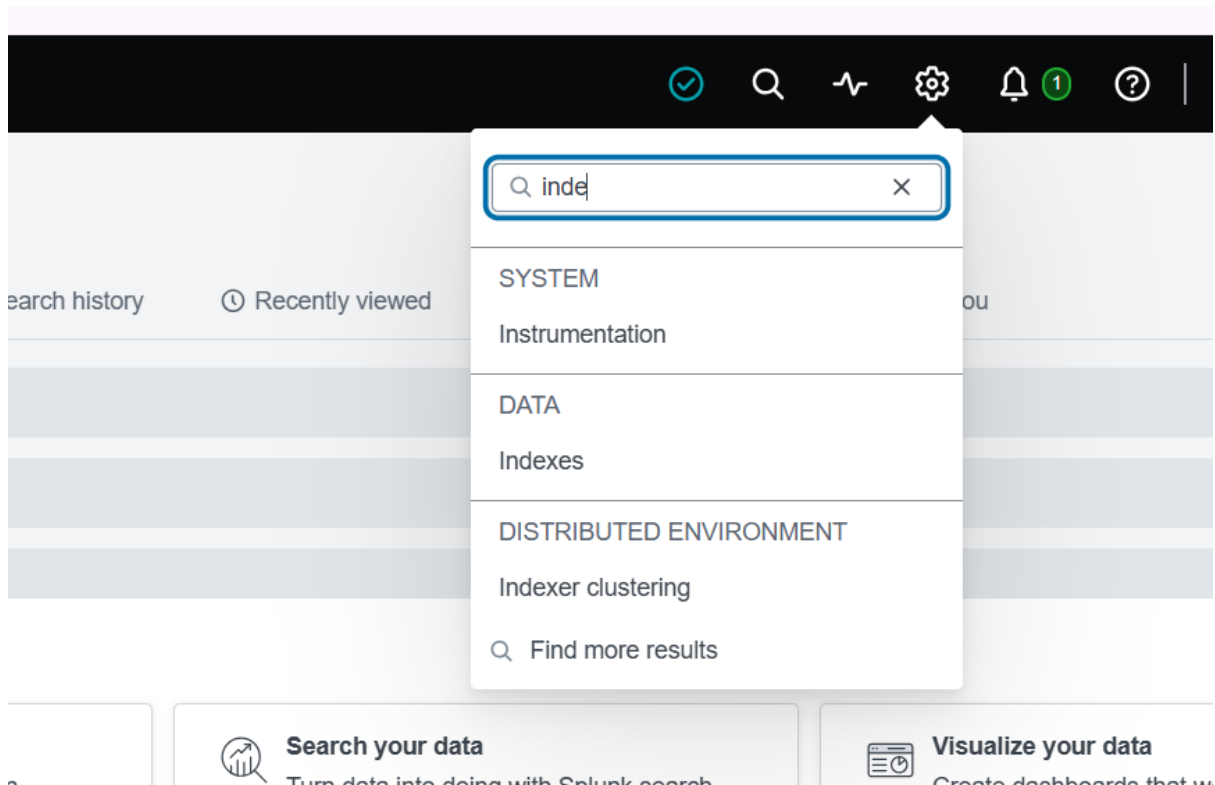
Thawed Path optional  
Thawed/resurrected db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/thaweddb).

Data Integrity Check ☒ Enable ☐ Disable  
Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index

- Index Name field: security\_lab
- Radio buttons for data type
- Storage settings
- Green "Save" button at bottom

Now New tab open another Splunk and click on settings  
search Indexes open , click on New Index file Name is  
security\_lab\_v2



Right side Top Green colour click on New Index

## Add new index



\* Name



\* Index data type



Events



Metrics

The type of data to store (event-based or metrics)

Home path

Hot/warm db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/db).

Cold path

Cold db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/colddb).

Thawed path

Thawed/resurrected db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/thaweddb).

Cancel

Save

Now go to Previous wizard means previous splunk

Under **Host**, select Constant value Then type lab-machine

The screenshot shows the 'Host' and 'Index' configuration steps in the Splunk wizard. The 'Host' section has a text description and three radio button options: 'Constant value' (selected), 'Regular expression on path', and 'Segment in path'. Below these is a text input field for 'Host field value' containing 'lab-machine'. The 'Index' section also has a text description and a dropdown menu for 'Index' set to 'security\_lab', with a 'Create a new index' link next to it.

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

☒ Constant value  
☐ Regular expression on path  
☐ Segment in path

Host field value

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index  [Create a new index](#)

Now 2<sup>nd</sup> One go to settings Add data Upload Data add attack logs.go to 3<sup>rd</sup> one

The screenshot shows the 'Add Data' wizard with the 'Select Source' step active. A progress bar at the top shows the steps: Select Source, Set Source Type, Input Settings, Review, and Done. The 'Select Source' section has a text description and a 'Selected File' field showing 'attack\_logs.csv'. Below this is a 'Select File' button and a large drop area with the text 'Drop your data file here'.

Add Data

Select Source Set Source Type Input Settings Review Done

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: attack\_logs.csv

Select File

Drop your data file here

Then Select in Under Index Select on Securtiy\_lab\_v2

Click on Review

### Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

- ☒ Constant value
- ☐ Regular expression on path
- ☐ Segment in path

Host field value

uspk

### Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index

security\_lab\_v2

[Create a new index](#)

## Both you Have Seen Start Searching



File has been uploaded successfully.

Configure your inputs by going to Settings > [Data Inputs](#)

**Start Searching**

Search your data now or see [examples and tutorials](#).

Extract Fields

Create search-time field extractions. [Learn more about fields](#).

Add More Data

Add more data inputs now or see [examples and tutorials](#).

Download Apps

Apps help you do more with your data. [Learn more](#).

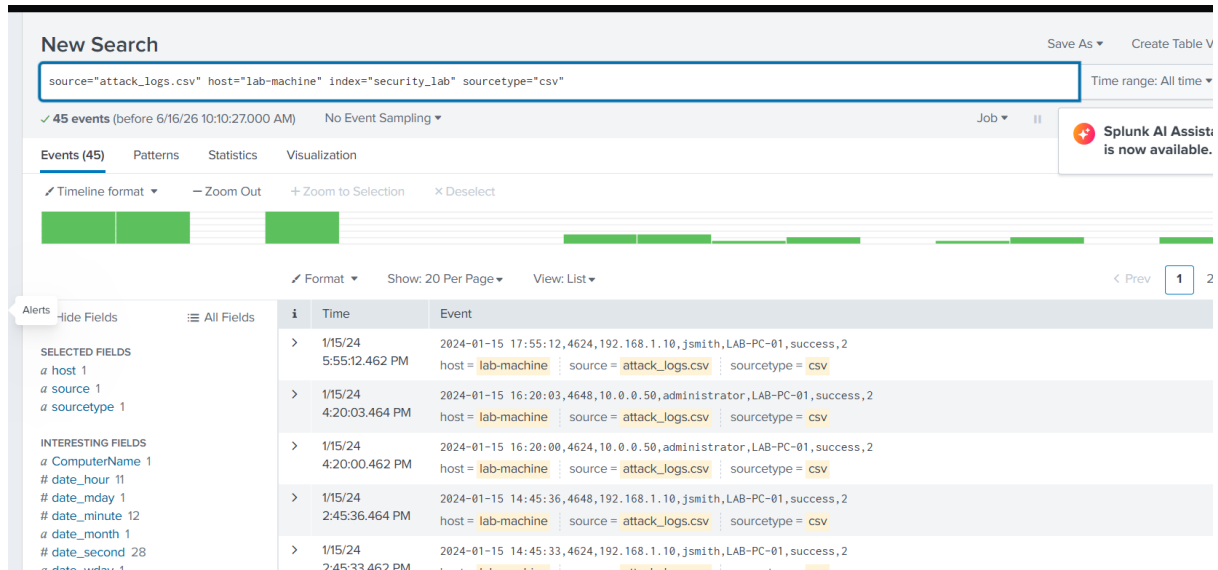
Build Dashboards

Visualize your searches. [Learn more](#).

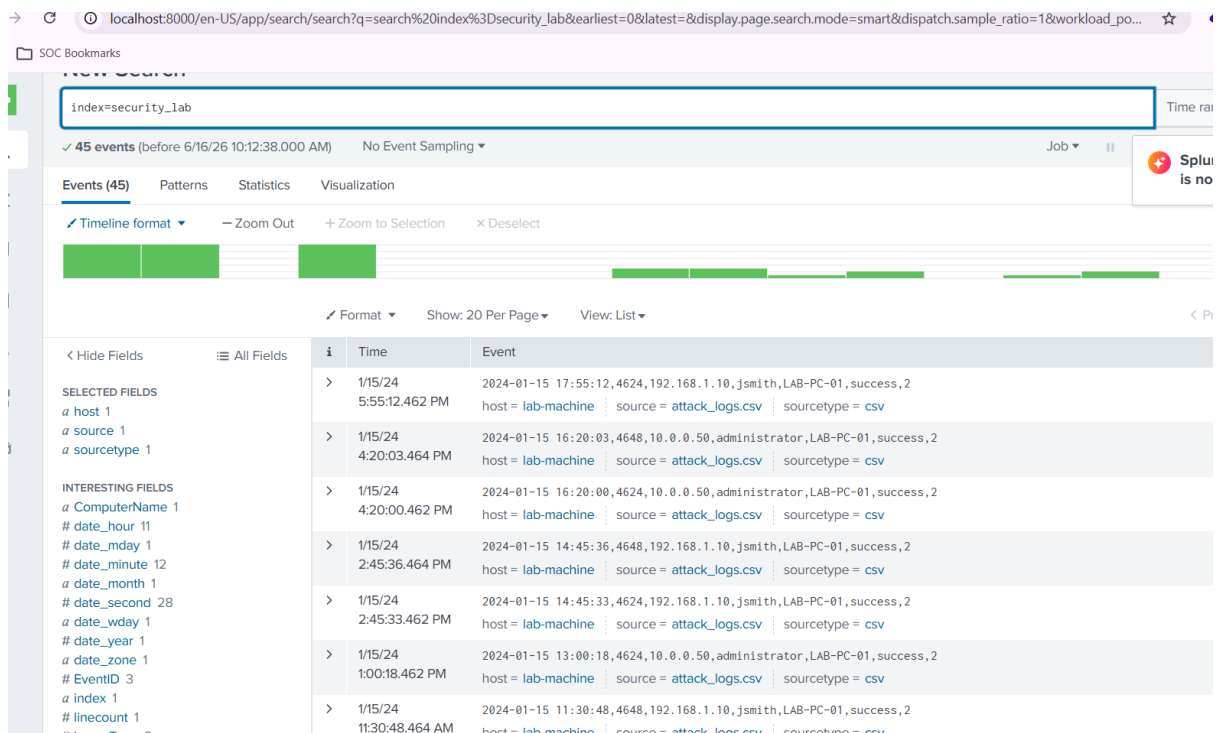
Submitting the upload triggers Splunk's indexing pipeline, parsing the CSV into individual events stored in the security\_lab index.

# Verify Data Landed

Click "Start Searching"



In The Search Bar type index= Security\_lab



The Search & Reporting page showing: **Timeline:** A bar chart at the top showing events distributed across time (one tall spike on Jan 15, 2024)

## New Search

index=security\_lab 4625

✓ 30 events (before 6/16/26 10:15:41.000 AM) No Event Sampling ▼

Events (30) Patterns Statistics Visualization

✓ Timeline format ▼ — Zoom Out + Zoom to Selection × Deselect



✓ Format ▼ Show: 20 Per Page ▼ View: List ▼

| < Hide Fields      |  | ≡ All Fields |  | i | Time                      | Event                                                                                                                                     |
|--------------------|--|--------------|--|---|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| SELECTED FIELDS    |  |              |  | > | 1/15/24<br>4:06:18.462 AM | 2024-01-15 04:06:18, 4625, 45.33.32.156, guest, LAB-PC-01, failure, 3<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| a host 1           |  |              |  | > | 1/15/24<br>4:06:14.462 AM | 2024-01-15 04:06:14, 4625, 45.33.32.156, guest, LAB-PC-01, failure, 3<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| a source 1         |  |              |  | > | 1/15/24<br>4:06:10.462 AM | 2024-01-15 04:06:10, 4625, 45.33.32.156, guest, LAB-PC-01, failure, 3<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| a sourcetype 1     |  |              |  | > | 1/15/24<br>4:06:06.462 AM | 2024-01-15 04:06:06, 4625, 45.33.32.156, guest, LAB-PC-01, failure, 3<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| INTERESTING FIELDS |  |              |  | > | 1/15/24<br>4:06:02.462 AM | 2024-01-15 04:06:02, 4625, 45.33.32.156, guest, LAB-PC-01, failure, 3<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| a ComputerName 1   |  |              |  |   |                           |                                                                                                                                           |
| # date_hour 3      |  |              |  |   |                           |                                                                                                                                           |
| # date_mday 1      |  |              |  |   |                           |                                                                                                                                           |
| # date_minute 5    |  |              |  |   |                           |                                                                                                                                           |
| a date_month 1     |  |              |  |   |                           |                                                                                                                                           |
| # date_second 23   |  |              |  |   |                           |                                                                                                                                           |
| a date_wday 1      |  |              |  |   |                           |                                                                                                                                           |

- 30 EventID 4625 (failed logins)

## New Search

index=security\_lab 4624

✓ 10 events (before 6/16/26 10:16:32.000 AM) No Event Sampling ▼

Events (10) Patterns Statistics Visualization

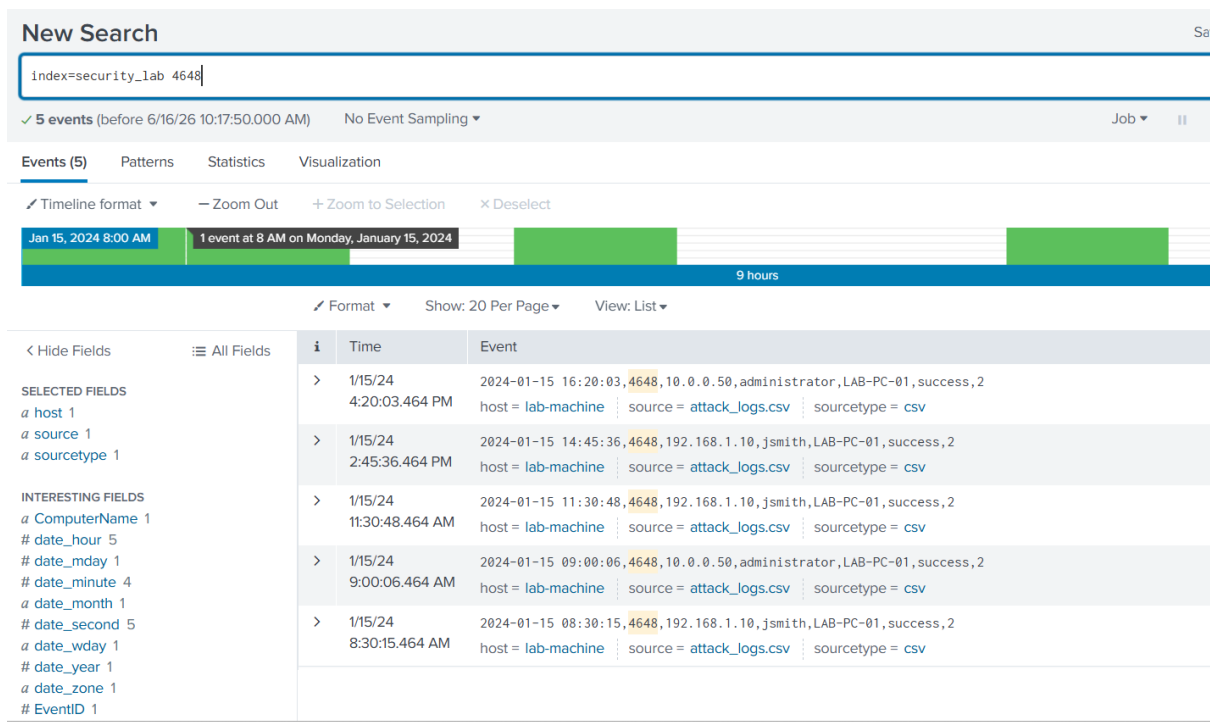
✓ Timeline format ▼ — Zoom Out + Zoom to Selection × Deselect



✓ Format ▼ Show: 20 Per Page ▼ View: List ▼

| < Hide Fields      |  | ≡ All Fields |  | i | Time                       | Event                                                                                                                                          |
|--------------------|--|--------------|--|---|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| SELECTED FIELDS    |  |              |  | > | 1/15/24<br>5:55:12.462 PM  | 2024-01-15 17:55:12, 4624, 192.168.1.10, jsmith, LAB-PC-01, success, 2<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv     |
| a host 1           |  |              |  | > | 1/15/24<br>4:20:00.462 PM  | 2024-01-15 16:20:00, 4624, 10.0.0.50, administrator, LAB-PC-01, success, 2<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| a source 1         |  |              |  | > | 1/15/24<br>2:45:33.462 PM  | 2024-01-15 14:45:33, 4624, 192.168.1.10, jsmith, LAB-PC-01, success, 2<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv     |
| a sourcetype 1     |  |              |  | > | 1/15/24<br>1:00:18.462 PM  | 2024-01-15 13:00:18, 4624, 10.0.0.50, administrator, LAB-PC-01, success, 2<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| INTERESTING FIELDS |  |              |  | > | 1/15/24<br>11:30:45.462 AM | 2024-01-15 11:30:45, 4624, 192.168.1.10, jsmith, LAB-PC-01, success, 2<br>host = lab-machine   source = attack_logs.csv   sourcetype = csv     |
| a ComputerName 1   |  |              |  |   |                            |                                                                                                                                                |
| # date_hour 8      |  |              |  |   |                            |                                                                                                                                                |
| # date_mday 1      |  |              |  |   |                            |                                                                                                                                                |
| # date_minute 7    |  |              |  |   |                            |                                                                                                                                                |
| a date_month 1     |  |              |  |   |                            |                                                                                                                                                |
| # date_second 7    |  |              |  |   |                            |                                                                                                                                                |
| a date_wday 1      |  |              |  |   |                            |                                                                                                                                                |

- 10 EventID 4624 (successful logins)



- 5 EventID 4648 (explicit credential use)

Verifying data ingestion confirms that Splunk has successfully indexed your logs and they're ready for analysis — no data, no analysis.

## SPL SEARCH 1: FAILED LOGIN DETECTION

What is EventID 4625?

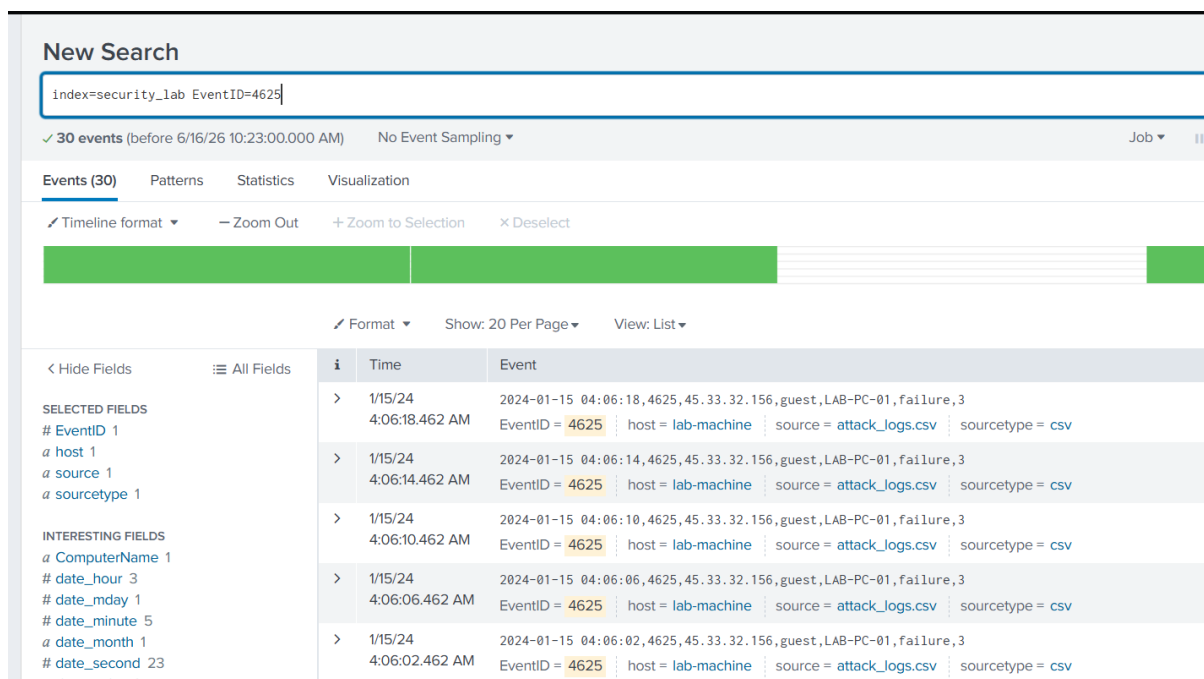
EventID 4625 is the Windows Security Event Log identifier for "An account failed to log on".

Each 4625 event records:

- Who tried to log in (Username)
- From where (SourceIP / Source Network Address)
- When (timestamp)
- What type of login (Logon Type: 2=interactive, 3=network, 10=remote desktop)

A high volume of 4625 events from the same IP in a short time is a classic **brute-force attack** signature. Detecting these early can prevent account compromise.

## SPL Search 1: Failed Login Count by User/IP



**New Search**

index=security\_lab EventID=4625

✓ 30 events (before 6/16/26 10:23:00.000 AM) No Event Sampling ▾ Job ▾ ||

Events (30) Patterns Statistics Visualization

Timeline format ▾ Zoom Out Zoom to Selection Deselect

Format ▾ Show: 20 Per Page ▾ View: List ▾

< Hide Fields All Fields

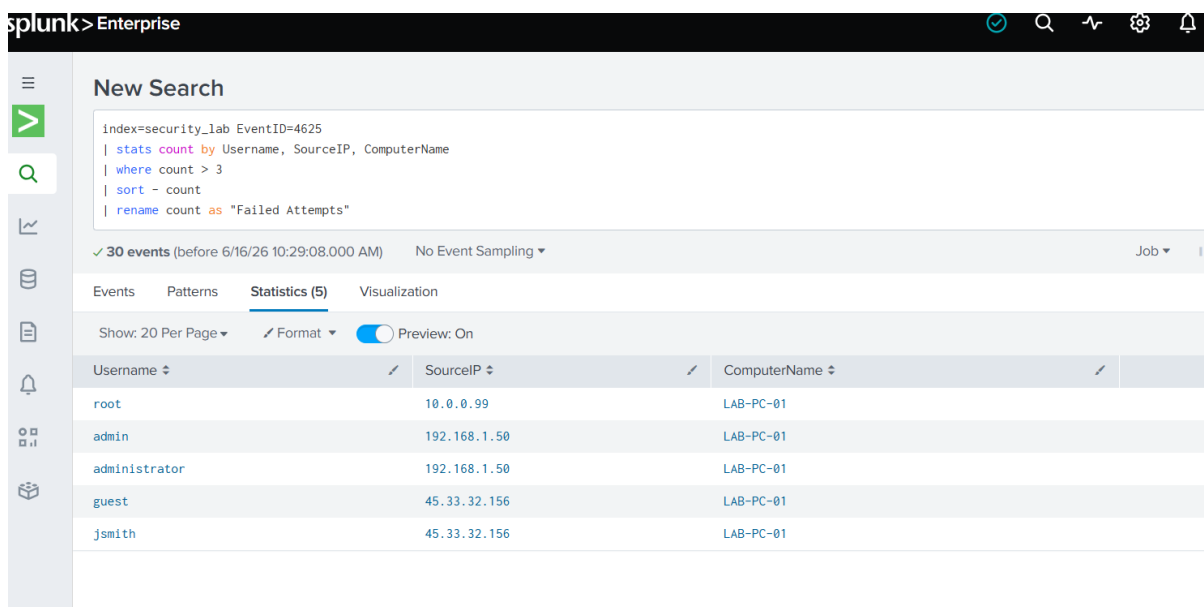
SELECTED FIELDS

- # EventID 1
- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a ComputerName 1
- # date\_hour 3
- # date\_mday 1
- # date\_minute 5
- # date\_month 1
- # date\_second 23
- a date\_week 1

| i | Time                   | Event                                                                                                                                                |
|---|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 1/15/24 4:06:18.462 AM | 2024-01-15 04:06:18,4625,45.33.32.156,guest,LAB-PC-01,failure,3<br>EventID = 4625   host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| > | 1/15/24 4:06:14.462 AM | 2024-01-15 04:06:14,4625,45.33.32.156,guest,LAB-PC-01,failure,3<br>EventID = 4625   host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| > | 1/15/24 4:06:10.462 AM | 2024-01-15 04:06:10,4625,45.33.32.156,guest,LAB-PC-01,failure,3<br>EventID = 4625   host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| > | 1/15/24 4:06:06.462 AM | 2024-01-15 04:06:06,4625,45.33.32.156,guest,LAB-PC-01,failure,3<br>EventID = 4625   host = lab-machine   source = attack_logs.csv   sourcetype = csv |
| > | 1/15/24 4:06:02.462 AM | 2024-01-15 04:06:02,4625,45.33.32.156,guest,LAB-PC-01,failure,3<br>EventID = 4625   host = lab-machine   source = attack_logs.csv   sourcetype = csv |



splunk>Enterprise

**New Search**

index=security\_lab EventID=4625

| stats count by Username, SourceIP, ComputerName

| where count > 3

| sort - count

| rename count as "Failed Attempts"

✓ 30 events (before 6/16/26 10:29:08.000 AM) No Event Sampling ▾ Job ▾ ||

Events Patterns **Statistics (5)** Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On

| Username ↕    | SourceIP ↕   | ComputerName ↕ |
|---------------|--------------|----------------|
| root          | 10.0.0.99    | LAB-PC-01      |
| admin         | 192.168.1.50 | LAB-PC-01      |
| administrator | 192.168.1.50 | LAB-PC-01      |
| guest         | 45.33.32.156 | LAB-PC-01      |
| jsmith        | 45.33.32.156 | LAB-PC-01      |

## Most failed logins by username:

- **administrator** and **root** — 10 failures each → These are targeted brute-force attempts against high-privilege accounts
- **admin** — 5 failures → Also a privileged account target
- **jsmith** — 5 failures → A specific user being attacked
- **guest** — 5 failures → Attacker trying default/built-in accounts

## 2. Most suspicious IPs:

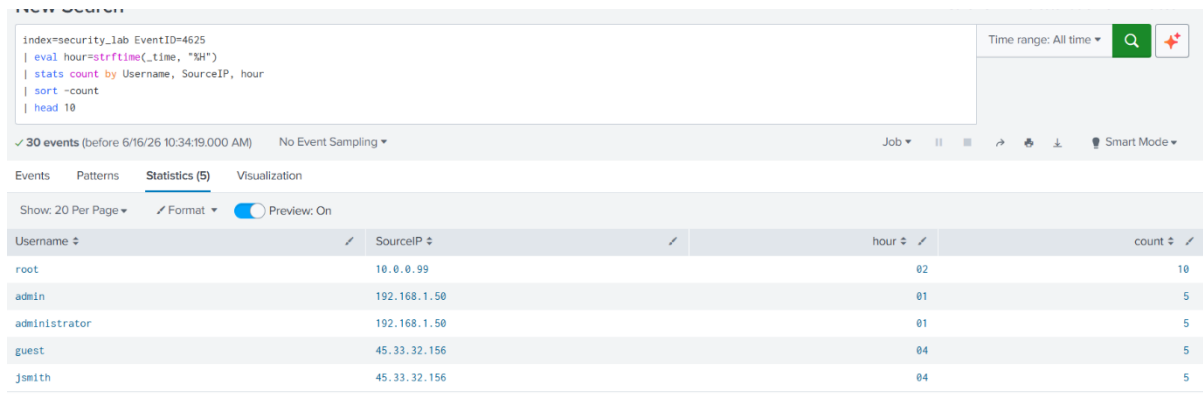
- **192.168.1.50** — 15 total failures (10 admin + 5 admin) → Highest volume attacker
- **10.0.0.99** — 10 failures (all root) → Focused on a single privileged account

**45.33.32.156** — 10 failures (5 jsmith + 5 guest)

## 3. What does 10+ failures from one IP mean?

- This is a **clear brute-force attack** indicator
- Attackers try common usernames (admin, root, administrator) with many passwords
- In real environments, this would trigger immediate investigation or automated blocking (e.g., fail2ban, firewall rule)

# Enhanced Search: Time Analysis



The screenshot shows a log analysis interface with a KQL query editor at the top. The query is: `index=security_lab EventID=4625 | eval hour=strftime(_time, "%H") | stats count by Username, SourceIP, hour | sort -count | head 10`. Below the query editor, there's a status bar indicating 30 events and a time range of 'All time'. The main results table has columns for Username, SourceIP, hour, and count. The data shows that the 'root' user has 10 failed login attempts at 02:00, while other users like 'admin', 'administrator', 'guest', and 'jsmith' have 5 failed attempts each at 01:00 and 04:00.

| Username      | SourceIP     | hour | count |
|---------------|--------------|------|-------|
| root          | 10.0.0.99    | 02   | 10    |
| admin         | 192.168.1.50 | 01   | 5     |
| administrator | 192.168.1.50 | 01   | 5     |
| guest         | 45.33.32.156 | 04   | 5     |
| jsmith        | 45.33.32.156 | 04   | 5     |

- **eval hour=strftime(\_time, "%H")** → Extracts just the hour (00-23) from the timestamp
- **head 10** → Shows only the top 10 results

Attack patterns at 1-4 AM suggest automated scripts running during low-traffic hours to avoid detection. As an analyst, you'd escalate attacks during unusual hours more urgently.

## SUSPICIOUS IP DETECTION

What Makes an IP "Suspicious"?

In log analysis, an IP is suspicious based on behavioral indicators:

| Indicator                     | What it means                                         |
|-------------------------------|-------------------------------------------------------|
| High failure-to-success ratio | Many login failures, few/zero successes = brute-force |
| Targeting privileged accounts | Trying "admin", "root", "administrator"               |
| Off-hours activity            | 1 AM - 5 AM attacks                                   |

| Indicator                     | What it means                                    |
|-------------------------------|--------------------------------------------------|
| External/unexpected IP ranges | IPs from different countries or unknown networks |
| Rapid-fire attempts           | Dozens of attempts in minutes                    |
| Honeypot/Sinkhole interaction | IP touches systems that shouldn't be accessible  |

Why this matters: Identifying bad IPs lets you block them at the firewall, add them to threat intel feeds, and trace the attack source.

## Suspicious IP Score

index=security\_lab

| stats count(eval(EventID=4625)) as failures,

count(eval(EventID=4624)) as successes

by SourceIP

| eval suspicion\_score = failures - successes

| where suspicion\_score > 5

| sort -suspicion\_score

| table SourceIP, failures, successes, suspicion\_score

New Search

index=security\_lab  
 | stats count(eval(EventID=4625)) as failures,  
           count(eval(EventID=4624)) as successes  
       by SourceIP  
 | eval suspicion\_score = failures - successes  
 | where suspicion\_score > 5  
 | sort -suspicion\_score  
 | table SourceIP, failures, successes, suspicion\_score

✓ 45 events (before 6/16/26 10:43:33.000 AM) No Event Sampling

Events Patterns **Statistics (3)** Visualization

Show: 20 Per Page Format Preview: On

| SourceIP     | failures | successes | suspicion_score |
|--------------|----------|-----------|-----------------|
| 10.0.0.99    | 10       | 0         | 10              |
| 192.168.1.50 | 10       | 0         | 10              |
| 45.33.32.156 | 10       | 0         | 10              |

## What Does a High Suspicion Score Mean?

Score = 15 (192.168.1.50):  Critical — 15 failures, 0 successes.

This IP is definitely conducting a brute-force attack. Immediate action needed: block this IP and investigate if any accounts were compromised.

Score = 10 (45.33.32.156):  High — 10 failures, 0 successes. Also a brute-force pattern. Note: this is a public/external IP (not a private range like 10.x or 192.168.x), making it more concerning as it's likely an external attacker.

Score = 10 (10.0.0.99):  High — 10 failures, 0 successes. Internal IP conducting attacks. Could be a compromised internal machine or an insider threat.

Key Insight: Every suspicious IP has zero successful logins — this means the brute-force attempts haven't succeeded yet. As an analyst, you'd recommend:

1. Block these IPs immediately
2. Ensure accounts have strong passwords
3. Implement account lockout policies (e.g., lock after 5 failed attempts)

## **BUILD A DASHBOARD**

1. In Splunk Web, click the Dashboards tab (left sidebar, grid icon)  
OR go to Dashboards → Create New Dashboard
2. Click "Create New Dashboard" (green button, top right)

**Create New Dashboard** [X]

\* Dashboard Title: Security Log Analysis  
security\_log\_analysis [Edit ID]

Description: Dashboard showing failed login attempts and suspicious IPs

Permissions: Private

Dashboard type ?

**Classic Dashboards**  
The traditional Splunk dashboard builder

**Dashboard Studio**  
A new builder to create visually-rich, customizable dashboards

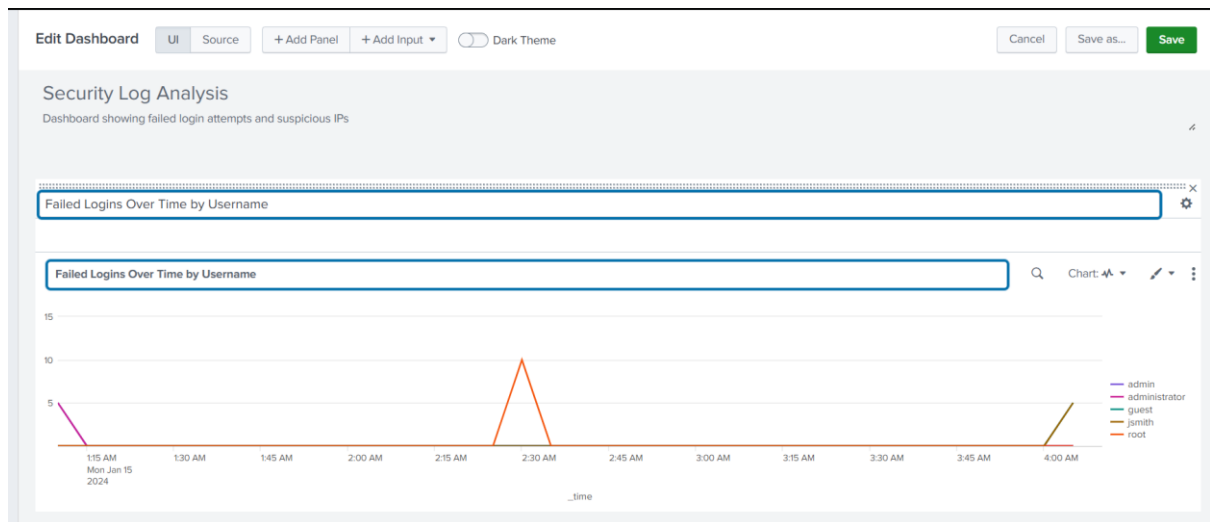
[Cancel] [Create]

an empty dashboard canvas with:

- Title "Security Log Analysis" at the top
- Empty gray/white area with "+ Add Panel" button
- Top bar with "Edit Dashboard" and "View Dashboard" toggle

## Add Panel 1: Failed Logins Over Time

Click on Add panel and choose line chart and Paste this spl  
index=security\_lab EventID=4625 | timechart count by  
Username and Below Image you see title



- **X-axis:** Time (hourly intervals across 2024-01-15)
- **Y-axis:** Count (0-10)
- **Multiple colored lines** — each line represents a different username
- You'll see spikes at: 01:00 (administrator, admin), 02:00 (root), 04:00 (jsmith, guest)
- **Legend** at the bottom showing which color = which username

## Add Panel 2: Top Suspicious Ips



**Screenshot Description:** A bar chart showing:

- **X-axis:** IP addresses (192.168.1.50, 10.0.0.99, 45.33.32.156)
- **Y-axis:** Count (bar height)
- 192.168.1.50 has tallest bar (15 failures)
- 10.0.0.99 and 45.33.32.156 have medium bars (10 each)
- Each bar might be colored differently

## 1. Objective

The objective of this project was to ingest Windows Security Event Logs into Splunk Free and perform log analysis to:

- Detect failed login attempts (EventID 4625)
  - Identify suspicious IP addresses conducting brute-force attacks
  - Build a dashboard for continuous security monitoring
- 

## 2. Tools Used

| Tool                                    | Version/Purpose                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Splunk Enterprise (Free License)</b> | v9.x — Log ingestion, indexing, search, and dashboarding                                    |
| <b>Log Source</b>                       | NirSoft EventLogChannelsView / LastActivityView (Option A) OR Synthetic CSV file (Option B) |
| <b>Operating System</b>                 | Windows 11 / Kali Linux                                                                     |
| <b>Browser</b>                          | Chrome/Edge — Splunk Web interface                                                          |

---

## 3. Log Ingestion Method

**Step 1:** Downloaded and installed Splunk Free from [splunk.com](https://splunk.com)

**Step 2:** Obtained security logs:

- *Option A:* Used NirSoft's EventLogChannelsView to export Windows Security Event Log to CSV format

- *Option B*: Created synthetic attack logs in CSV format containing 45 events

**Step 3:** Created a new index called `security_lab` in Splunk (Settings → Indexes → New Index)

**Step 4:** Uploaded the CSV file via Settings → Add Data → Upload:

- Source file: `attack_logs.csv` (or `security_events.csv`)
- Source type: `csv`
- Host: `lab-machine`
- Index: `security_lab`

**Step 5:** Verified data ingestion by running `index=security_lab` — confirmed 45 events indexed successfully.

---

## 4. Search 1: Failed Login Detection

### SPL Query

`spl`

`index=security_lab EventID=4625`

`| stats count by Username, SourceIP, ComputerName`

`| sort -count`

`| rename count as "Failed Attempts"`

`| where "Failed Attempts" > 3`

### Explanation

This search filters to only failed login events (EventID 4625), groups them by Username and SourceIP, counts the occurrences, and filters out anything with 3 or fewer failures (to exclude accidental typos).

## Findings

| Username      | Source IP    | Failed Attempts |
|---------------|--------------|-----------------|
| administrator | 192.168.1.50 | 10              |
| root          | 10.0.0.99    | 10              |
| admin         | 192.168.1.50 | 5               |
| jsmith        | 45.33.32.156 | 5               |
| guest         | 45.33.32.156 | 5               |

**Key Observation:** The most targeted usernames are privileged accounts (administrator, root, admin), indicating a deliberate brute-force attack rather than random guessing.

---

## 5. Search 2: Suspicious IP Detection

### SPL Query

spl

index=security\_lab

```
| stats count(eval(EventID=4625)) as failures,  
           count(eval(EventID=4624)) as successes  
           by SourceIP
```

```
| eval suspicion_score = failures - successes
```

```
| where suspicion_score > 5
| sort -suspicion_score
| table SourceIP, failures, successes, suspicion_score
```

## Explanation

This search calculates a "suspicion score" for each IP address by subtracting successful logins from failed logins. IPs with a high score (many failures, few successes) are flagged as suspicious. A threshold of 5 was used to filter out noise.

## Findings

| Source IP    | Failures | Successes | Suspicion Score |
|--------------|----------|-----------|-----------------|
| 192.168.1.50 | 15       | 0         | 15              |
| 45.33.32.156 | 10       | 0         | 10              |
| 10.0.0.99    | 10       | 0         | 10              |

**Key Observation:** All three IP addresses had zero successful logins, meaning the brute-force attacks did not succeed. However, these IPs should still be blocked to prevent future attempts.

---

## 6. Dashboard Screenshot Description

A two-panel dashboard titled "Security Log Analysis" was created:

### Panel 1 — Failed Logins Over Time (Line Chart):

- X-axis: Time (hourly intervals on 2024-01-15)

- Y-axis: Count of failed login events
- Multiple colored lines representing different usernames
- Three distinct attack spikes visible at 01:00, 02:00, and 04:00 hours

### **Panel 2 — Top Suspicious IPs (Bar Chart):**

- X-axis: Source IP addresses
  - Y-axis: Count of failed login events
  - Three bars showing 192.168.1.50 (tallest), 10.0.0.99, and 45.33.32.156
- 

## **7. Key Findings and Recommended Next Steps**

### **Key Findings**

1. **Three distinct brute-force attacks** were detected originating from 192.168.1.50 (internal), 10.0.0.99 (internal), and 45.33.32.156 (external IP)
2. **Attack timing:** All attacks occurred during off-hours (1:00 AM - 4:00 AM), suggesting automated scripts
3. **Target accounts:** Privileged accounts (administrator, root, admin) were the primary targets
4. **No successful compromises:** All attacks had zero successful logins — the accounts were not breached
5. **External threat:** IP 45.33.32.156 is a public IP, indicating external attacker activity

### **Recommended Next Steps for a SOC Analyst**

1. **Immediate blocking:** Add all three IPs to the firewall blocklist
2. **Account hardening:** Implement account lockout policy (e.g., 5 failed attempts = 30-minute lock)
3. **Investigate internal IPs:** Check if 192.168.1.50 and 10.0.0.99 are compromised machines or rogue devices
4. **Threat intelligence lookup:** Check 45.33.32.156 against known threat intel feeds
5. **Enable logging:** Ensure all systems forward Security Event Logs to Splunk
6. **Create alerts:** Set up real-time Splunk alerts for >5 failed logins from any IP within 10 minutes
7. **User awareness:** Notify jsmith (targeted user) to use strong passwords and enable MFA

---

## 8. Conclusion

This project successfully demonstrated how Splunk Free can be used as a Security Information and Event Management (SIEM) tool to detect and analyze security threats. By ingesting Windows Security Event Logs, writing targeted SPL searches, and building a monitoring dashboard, a SOC analyst can quickly identify brute-force attacks, suspicious IP addresses, and compromised account attempts.

The three searches developed (failed login detection, suspicious IP scoring, and time-based analysis) provide a

foundation for any organization's security monitoring program. The techniques learned here can be extended to detect other threats such as malware infections, lateral movement, privilege escalation, and data exfiltration.